Formal Analysis of ZAP-WS Per-Stream Independence: Tail-Latency Independence and Throughput Composition

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2026

Abstract

We formalize the per-stream independence claim of ZAP-WS. Under a standard packet-erasure channel model, we prove (Theorem 1) that the arrival-time distribution of any frame on stream s_i is statistically independent of the existence of, or losses affecting, frames on stream s_j for $j \neq i$. We prove (Theorem 2) that aggregate throughput across N streams is bounded by the underlying link capacity but is otherwise independent of stream count, modulo per-stream framing overhead. We prove (Lemma) the standard Reed–Solomon FEC overhead bound: a $(k+m, k)$ code recovers from up to m erasures with bandwidth overhead exactly m/k. The proofs are constructive enough to guide a conformant implementation.

1 Model

1.1 Connections, packets, streams

A ZAP-WS connection C runs between two peers over an erasure-channel substrate (UDP, raw IP, etc.). The connection carries a sequence of *packets* $\pi_1, \pi_2, \ldots$ Each packet is transmitted as a single substrate datagram and either arrives intact or is lost. Each packet contains one or more *frames*, each frame belonging to exactly one of N logical *streams* $s_1, \ldots, s_N$.

Assumption 1 (Bernoulli erasure)**.** *Each transmitted packet is independently lost with probability p ($0 \leq p < 1$) and otherwise arrives after a propagation delay D. Different packets are lost independently.*

Assumption 2 (Single-stream packing)**.** *Each packet contains frames belonging to a single stream. (This assumption strengthens the claim. A weaker, multi-stream-per-packet model and its consequences are discussed in Section 5.)*

Assumption 3 (Independent recovery)**.** *Each stream uses either independent retransmission with one round-trip recovery delay R, or independent (k, m) Reed–Solomon FEC. The recovery decision and parameters for stream s_i are made without reference to any other stream.*

1.2 Latency random variable

For frame f of stream s_i, let

$$L(s_i, f) := T_{\text{deliver}}(s_i, f) - T_{\text{send}}(s_i, f)$$

be the application-to-application latency. We are interested in the distribution of $L(s_i, f)$ and how it depends on traffic on other streams.

2 Theorem 1: Tail-Latency Independence

Theorem 4 (Tail-Latency Independence)**.** *Under Assumptions 1–3, for any two distinct streams $s_i \neq s_j$ on the same ZAP-WS connection, any frame f on s_i, and any threshold $t \geq 0$,*

$$\Pr\big[L(s_i, f) > t \,\big|\, \mathcal{F}_j\big] = \Pr\big[L(s_i, f) > t\big],$$

where $\mathcal{F}_j$ is the σ-algebra generated by all events (send, loss, recovery) on stream s_j.

Proof. We show that $L(s_i, f)$ depends only on the substrate-level packet events affecting frame f, which are independent of substrate-level events on packets carrying s_j frames.

Step 1: $L(s_i, f)$ as a function of s_i packet events. By Assumption 2, every packet that carries any byte of frame f carries only s_i data. Let $\Pi_i(f) \subseteq \{\pi_1, \pi_2, \ldots\}$ be the (finite) set of substrate packets that contain either f's bytes or s_i's FEC parity for f's FEC group, or any s_i retransmission packet covering f. The arrival event of f at the application is determined by:

(a) Which packets in $\Pi_i(f)$ are lost (Bernoulli, by Assumption 1).

(b) Whether any subset of size $\geq k$ of the FEC group survives, if FEC is used (deterministic given losses).

(c) Otherwise, retransmission timing for the lost subset (deterministic given the recovery rule of Assumption 3, which references only s_i).

Hence $L(s_i, f)$ is a measurable function of the joint loss outcome on $\Pi_i(f)$ alone.

Step 2: $\Pi_i(f) \cap \Pi_j(\cdot) = \emptyset$. For any frame g on s_j, let $\Pi_j(g)$ be the analogously defined packet set. By Assumption 2, packets in $\Pi_i(\cdot)$ contain only s_i frames and packets in $\Pi_j(\cdot)$ contain only s_j frames; since each substrate packet is generated as a single ZAP-WS frame plus header, the packet sets are disjoint.

Step 3: Independence of disjoint Bernoulli events. By Assumption 1, the loss outcomes on disjoint sets of packets are independent. Therefore the loss outcome on $\Pi_i(f)$ is independent of the loss outcomes on every $\Pi_j(g)$, and hence independent of $\mathcal{F}_j$.

Step 4: Conclude. By Step 1, $L(s_i, f)$ is a function of a random variable independent of $\mathcal{F}_j$. Hence

$$\Pr[L(s_i, f) > t \mid \mathcal{F}_j] = \Pr[L(s_i, f) > t]. \qquad \square$$

Corollary 5 (No HoL blocking)**.** *For ZAP-WS, an arbitrarily long stall on stream s_j cannot increase the tail latency $\Pr[L(s_i, f) > t]$ for any frame on stream s_i.*

Contrast with TCP-multiplexed protocols. For WebSocket and HTTP/2, the analogue of $\Pi_i(f)$ contains every TCP segment whose sequence numbers precede f's last byte in the TCP byte stream, including segments carrying s_j frames. Thus $\Pi_i(f) \cap \Pi_j(\cdot) \neq \emptyset$ in general, Step 2 fails, and conditional independence does not hold.

3 Theorem 2: Throughput Composition

Theorem 6 (Throughput Composition)**.** *Let Θ_i be the long-run achieved throughput of stream s_i under saturating offered load and infinite per-stream window. Under Assumptions 1–3, for any partition of $\{1, \ldots, N\}$,*

$$\sum_{i=1}^{N} \Theta_i \;\leq\; B \cdot (1-p) - O_h(N),$$

where B is the link capacity, p is the packet loss probability, and $O_h(N) = N \cdot \omega$ for a per-stream framing overhead ω (headers plus FEC parity in the FEC mode). Furthermore, the achievable $(\Theta_1, \ldots, \Theta_N)$ region is convex and the per-stream throughput Θ_i depends on other streams only *through the shared capacity constraint.*

Proof sketch. The substrate is a single physical link of capacity B. Each packet either carries a stream's payload (counts toward that stream's throughput) or the framing/FEC overhead. Per-stream framing overhead ω is additive in N because each stream contributes a constant header rate. Conservation of bandwidth gives the upper bound. The achievable region is convex by time-sharing arguments standard in network capacity analysis. Per-stream loss recovery (Assumption 3) implies Θ_i's loss-recovery dynamics depend only on s_i's own losses; hence apart from sharing B, the streams are decoupled. □

Corollary 7. *Doubling the number of streams (with fixed aggregate offered load) does not change tail latency on any individual stream (Theorem 4) and reduces aggregate goodput only by ω per added stream.*

4 Lemma: Reed–Solomon FEC Overhead Bound

Lemma 8 (FEC overhead bound)**.** *Let a stream use a systematic $(k+m, k)$ Reed–Solomon code over $\mathrm{GF}(2^q)$, $q \geq \lceil \log_2(k+m) \rceil$, applied per FEC group of k data frames. Then:*

(i) Any subset of k of the $k+m$ transmitted frames is sufficient to reconstruct the k original data frames.

(ii) The bandwidth overhead of FEC is exactly m/k.

Proof. **(i)** A Reed–Solomon code is a maximum-distance separable (MDS) code: its minimum distance is $m+1$ and any k symbols of the codeword uniquely determine the message [1]. Standard.

(ii) For each k data frames the encoder emits $k+m$ codeword frames. The ratio of transmitted bytes to data bytes is $(k+m)/k = 1 + m/k$, so the overhead above the data rate is m/k, exactly. □ □

Corollary 9 (Recovery without retransmission)**.** *With (k, m) FEC and per-packet erasure probability p, the probability that a FEC group fails to reconstruct (i.e., suffers $> m$ erasures out of $k+m$) is*

$$\Pr[\mathit{FEC\ fail}] = \sum_{j=m+1}^{k+m} \binom{k+m}{j} p^j (1-p)^{k+m-j}.$$

For p small and m chosen so that $(k+m)p < m$, this is exponentially small in m (Chernoff). Tuning m thus exchanges bandwidth for tail latency: each unit increase in m exponentially reduces retransmission probability and thereby reduces $\Pr[L(s_i, f) > D + R]$ exponentially.

5 Multi-Stream-Per-Packet Packing

In practice, ZAP-WS implementations may pack frames from multiple streams into a single substrate packet to amortize header overhead. We quantify the cost.

Proposition 10 (Soft HoL under co-packing)**.** *Suppose with probability α a substrate packet carries frames from two or more streams. Then the conditional independence statement of Theorem 4 holds with conditional dependence contribution at most α. Specifically,*

$$\left|\Pr[L(s_i, f) > t \mid \mathcal{F}_j] - \Pr[L(s_i, f) > t]\right| \leq \alpha.$$

Proof sketch. The argument of Theorem 1 fails only on the event E that a packet in $\Pi_i(f)$ also belongs to $\Pi_j(\cdot)$ for some j. By construction $\Pr[E] \leq \alpha$. Outside E, the conditional probability equals the unconditional one. By total probability the difference is bounded by α. □

Corollary 11. *A conformant implementation that wishes to preserve the bound of Theorem 4 exactly should refuse to pack frames from multiple streams in one substrate packet, accepting the per-stream header overhead. Implementations that prioritize throughput may set $\alpha > 0$ but must document the resulting weakening.*

6 Liveness

Theorem 12 (Per-stream liveness)**.** *Under Assumptions 1–3, suppose stream s_i has retransmission enabled with bounded RTO. Then for any frame f on s_i, $\Pr[L(s_i, f) < \infty] = 1$ regardless of the behaviour of any other stream s_j, $j \neq i$.*

Proof. Each retransmission of f is an independent Bernoulli trial of success probability $1-p > 0$. By Borel–Cantelli, infinitely many trials yield a success almost surely, hence f arrives in finite time a.s. The behaviour of s_j is irrelevant by the disjoint-packet-set argument of Theorem 4. □

7 Discussion

The proofs above identify three load-bearing assumptions:

1. **Erasure independence (Assumption 1).** Real networks exhibit bursty loss; the i.i.d. assumption is a useful first approximation but burstiness can correlate losses across streams that share queues. This is a property of the network, not of ZAP-WS.

2. **Single-stream packing (Assumption 2).** The cleanest independence holds when each substrate packet belongs to a single stream. Implementations may relax this for throughput; Proposition 10 bounds the cost.

3. **Independent recovery (Assumption 3).** Per-stream loss detection and either per-stream retransmission or per-stream FEC. This is exactly the design choice that distinguishes QUIC and ZAP-WS from TCP.

A stream substrate that violates any of these (e.g., a single shared ARQ window across all streams, as in TCP) re-introduces head-of-line blocking through the shared component.

8 Conclusion

Theorem 4 formalizes the core ZAP-WS claim: per-stream tail latency is statistically independent of other streams. Theorem 6 establishes that throughput composes up to shared link capacity. Lemma 8 bounds the FEC bandwidth cost exactly. Together, these results justify the engineering claim that ZAP-WS provides per-stream flow control with bounded, predictable tail latency under loss, on top of post-quantum mutual authentication inherited from the ZAP handshake.

References

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